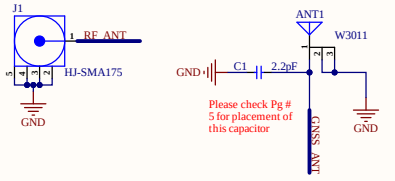
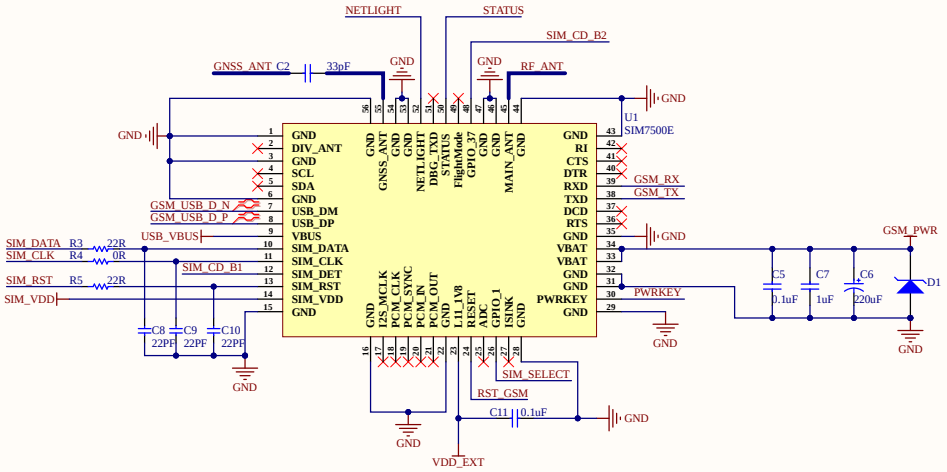
ANTENNA



Ground cleared under antenna, clearance area 4.00 x 4.25 mm
Feed line should be designed to match 50 ohm characteristic impedance, depending on PWB material and thickness.
Recommended test board layout for electrical characteristic measurement, test board outline size 80 x 37 mm.
For more info check datasheet

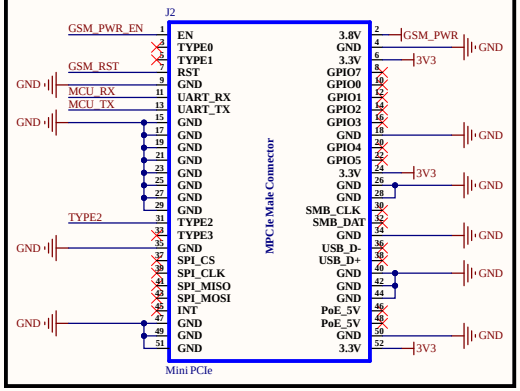
SIM7500E



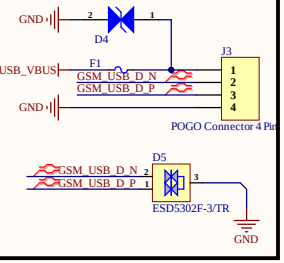
SIM Card Design Guide
SIM card signal could be interferenced by some high frequency signal, it is strongly recommendedtofollowthese guidelines while designing:
- SIM card holder should be far away from main antenna
- SIM traces should keep away from RF lines, VBAT and high-speed signal lines
- The traces should be as short as possible
- Keep SIM card holder's GND connect to main ground directly
- Shielding the SIM card signal by ground well
- Recommended to place a 100nF capacitor on SIM_VDD line and keep close to the SIMcardholder
- Add some TVS which parasitic capacitance should not exceed 50pF
- Add 22Ω resistor to (SIM_RST/SIM_CLK/SIM_DATA) signal could enhance ESDprotection Mount 22pF capacitor to (SIM_RST/SIM_CLK/SIM_DATA) signal if having RF signal interference

Datasheet Pg 31

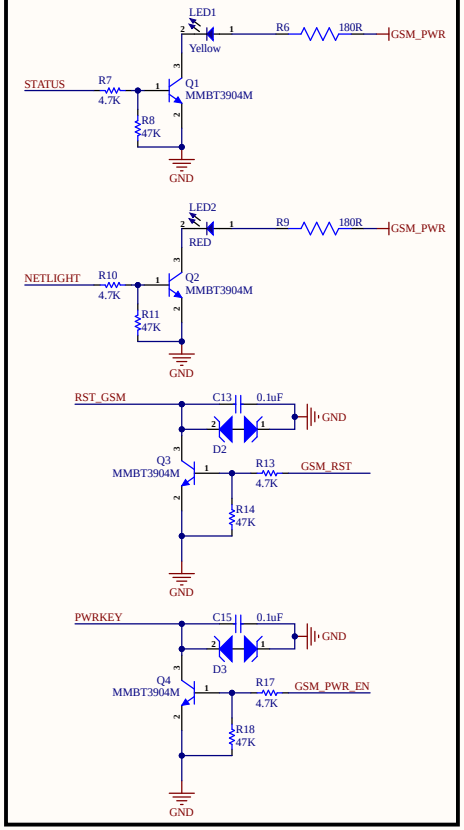
Mini PCIe Connector Male



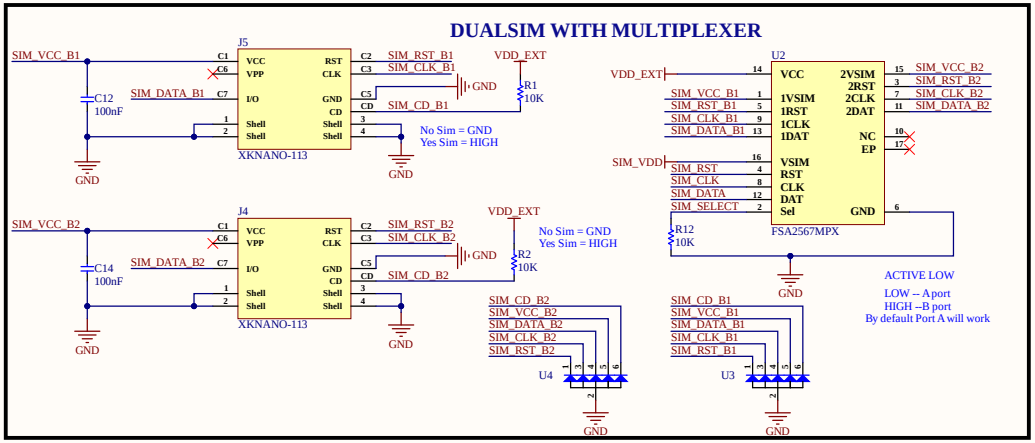
USB POGO for Updates



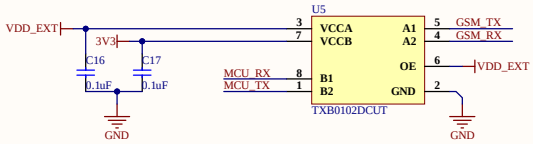
GSM_COMMANDS



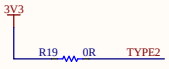
DUALSIM WITH MULTIPLEXER



Level Shifter for GSM Serial



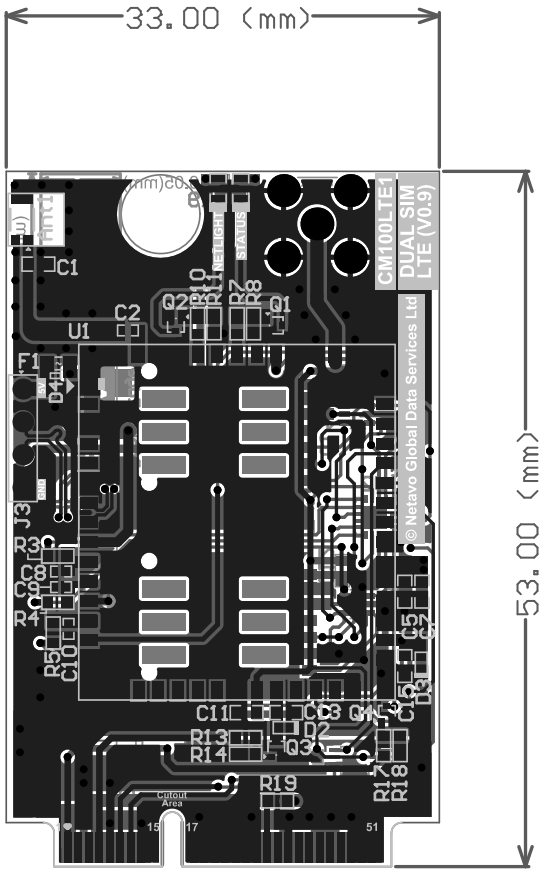
COMMS MODULE INDICATION



LTE Module indication is 0100
Type 0,1,3 are pull-down on baseboard

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Date: 10/13/2023	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 1



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.59mil	3.8	
1	Top Layer		1.38mil		
	Dielectric 1	FR4	35.43mil	4.8	
2	Bottom Layer		1.38mil		
	Bottom Solder	Solder Resist	0.59mil	3.8	
	Bottom Overlay				